class11: AlphaFold

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Table of contents

Background 1
Alpha Fold 1
EBI AlphaFold database 2
Running the AlphaFold 2
Interpreting Results 3
Now lets plot the pLDDT values across models 5
Predicted Alignment Error (PAE) for Domains 10
Plot N by N (number of residues) 11
Residue Conservation from Alignment File 13

Background

We saw last day that the main repository for biomolecular structure (the PDB database) only has ~250,000 entries.

UniProtKB (the main protein sequence database) has over 200 million entries!

Alpha Fold

In this hands-on session we will utilize AlphaFold to predict protein structure from sequence (Jumper et al. 2021).

Without the aid of such approaches, it can take years of expensive laboratory work to determine the structure of just one protein. With AlphaFold we can now accurately compute a typical protein structure in as little as ten minutes.

EBI AlphaFold database

The EBI AlphaFold database contains lots of computed structure models. It is increasing likely that the structure you are interested in is already in this database at https://alphafold.ebi.ac.uk/

There are 3 major outputs from AlphaFold:

1. A model of structure in **PDB** format.
2. A **pLDDT score** that tells us how confident the model is for a given residue in your protein (high values are good, above 70).
3. A **PAE score** that tells us about protein packing quality.

If you can't find a matching entry for the sequence you are interested in in AFDB, you can run AlphaFold yourself…

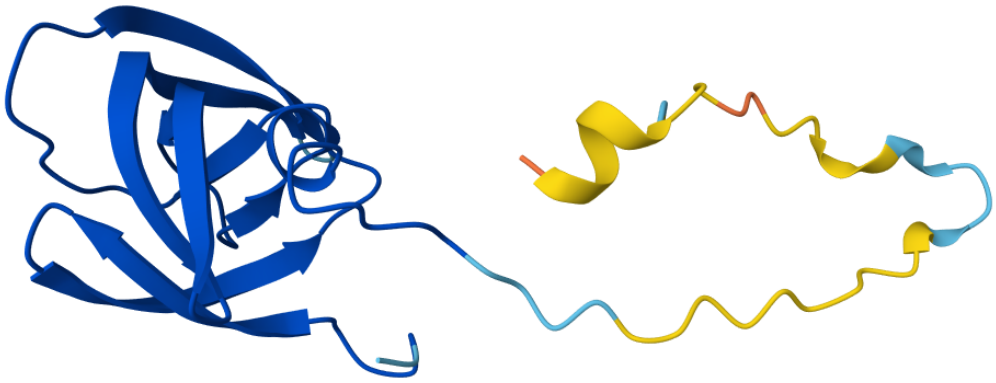


Figure 1: AFDB Entry of Peptidase A2 domain-containing protein

Running the AlphaFold

We will use ColabFold to run AlphaFold on our sequence https://colab.research.google.com/github/sokrypton/ColabFold/blob/main/AlphaFold2.ipynb

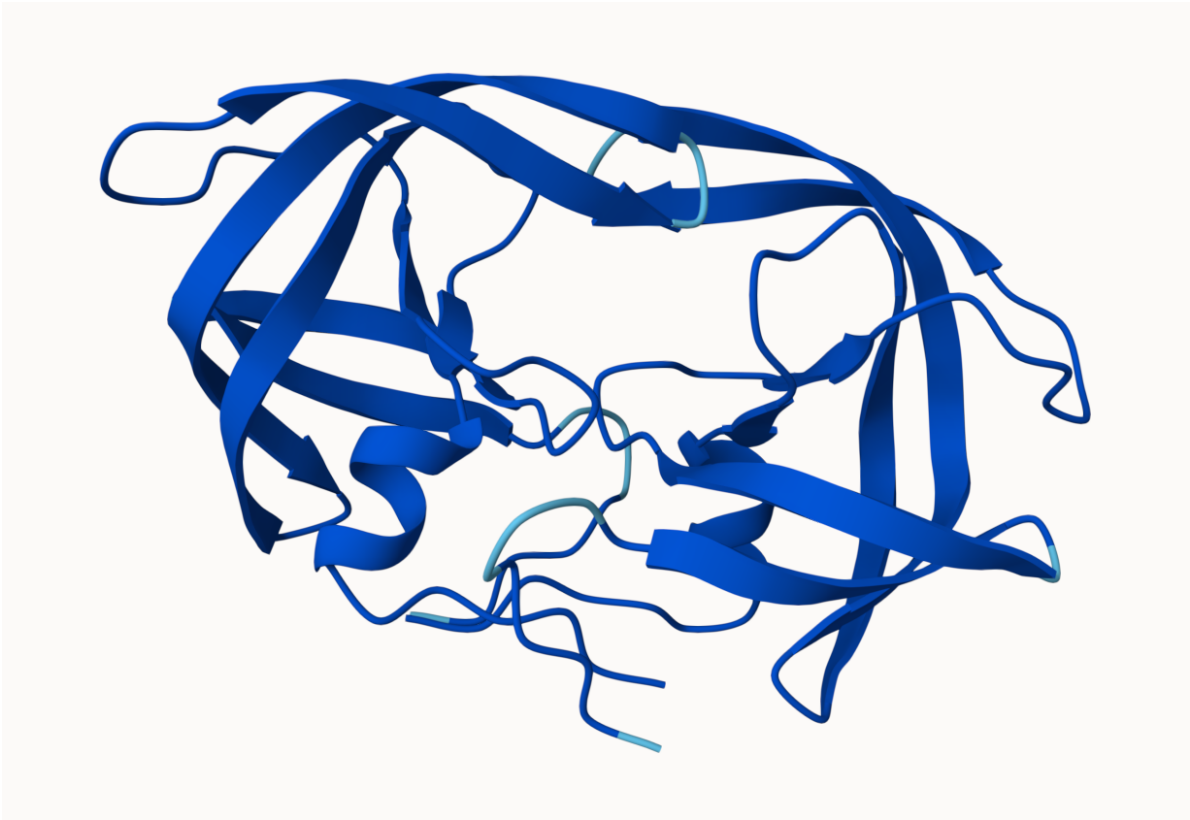


Figure 2: Estimate of how well predicted structure converges with MSA data and PDB structure, colored by IDDT("predicted local distance difference test")

Interpreting Results

Custom Analysis of resulting models

We can read all the AlphaFold results into R and do more quantitative analysis than just viewing the structures in Mol-star:

Read all the PDB models:

```
library(bio3d)
```

```
# Change this for YOUR results dir name
results_dir <- "HIVpr_23119/"

# File names for all PDB models
pdb_files <- list.files(path=results_dir,
```

```
                        pattern="*.pdb",
                        full.names = TRUE)
pdbs <- pdbaln(pdb_files, fit=TRUE, exefile="msa")
```

```
Reading PDB files:
HIVpr_23119//HIVpr_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_000.pdb
HIVpr_23119//HIVpr_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_000.pdb
HIVpr_23119//HIVpr_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_000.pdb
HIVpr_23119//HIVpr_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_000.pdb
HIVpr_23119//HIVpr_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000.pdb
.....

Extracting sequences

pdb/seq: 1   name: HIVpr_23119//HIVpr_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4
pdb/seq: 2   name: HIVpr_23119//HIVpr_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1
pdb/seq: 3   name: HIVpr_23119//HIVpr_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5
pdb/seq: 4   name: HIVpr_23119//HIVpr_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2
pdb/seq: 5   name: HIVpr_23119//HIVpr_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3
```

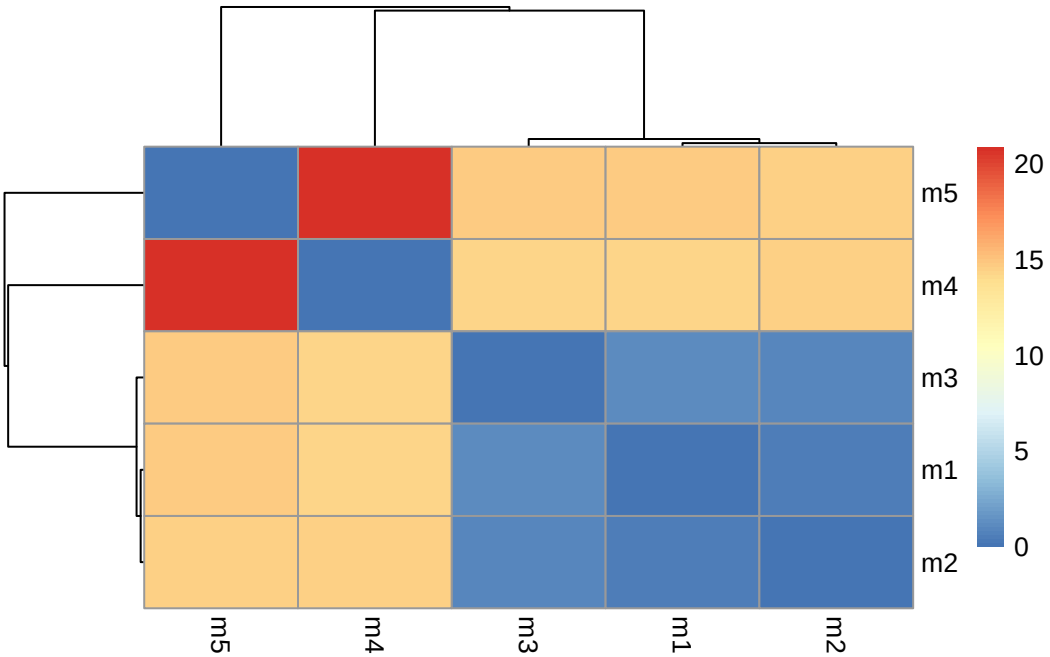
```
#library(bio3dview)
#view.pdbs(pdbs)
```

How similar or different are my models?

```
rd <- rmsd(pdbs)
```

```
Warning in rmsd(pdbs): No indices provided, using the 198 non NA positions
```

```
library(pheatmap)
colnames(rd) <- paste0("m",1:5)
rownames(rd) <- paste0("m",1:5)
pheatmap(rd)
```



Now lets plot the pLDDT values across models

```
pdb <- read.pdb("1hsg")
```

```
  Note: Accessing on-line PDB file
```

Find the "rigid core" across all models:

```
core <- core.find(pdbs)
```

```
 core size 197 of 198  vol = 9885.822
 core size 196 of 198  vol = 6896.71
 core size 195 of 198  vol = 1337.847
 core size 194 of 198  vol = 1040.67
 core size 193 of 198  vol = 951.857
 core size 192 of 198  vol = 899.083
 core size 191 of 198  vol = 834.732
 core size 190 of 198  vol = 771.338
 core size 189 of 198  vol = 733.065
```

```
core size 188 of 198  vol = 697.28
core size 187 of 198  vol = 659.742
core size 186 of 198  vol = 625.273
core size 185 of 198  vol = 589.541
core size 184 of 198  vol = 568.253
core size 183 of 198  vol = 545.015
core size 182 of 198  vol = 512.889
core size 181 of 198  vol = 490.723
core size 180 of 198  vol = 470.266
core size 179 of 198  vol = 450.731
core size 178 of 198  vol = 434.735
core size 177 of 198  vol = 420.337
core size 176 of 198  vol = 406.658
core size 175 of 198  vol = 393.334
core size 174 of 198  vol = 382.395
core size 173 of 198  vol = 372.858
core size 172 of 198  vol = 356.994
core size 171 of 198  vol = 346.567
core size 170 of 198  vol = 337.446
core size 169 of 198  vol = 326.659
core size 168 of 198  vol = 314.95
core size 167 of 198  vol = 304.127
core size 166 of 198  vol = 294.552
core size 165 of 198  vol = 285.648
core size 164 of 198  vol = 278.884
core size 163 of 198  vol = 266.765
core size 162 of 198  vol = 258.994
core size 161 of 198  vol = 247.723
core size 160 of 198  vol = 239.84
core size 159 of 198  vol = 234.963
core size 158 of 198  vol = 230.062
core size 157 of 198  vol = 221.985
core size 156 of 198  vol = 215.62
core size 155 of 198  vol = 206.793
core size 154 of 198  vol = 196.984
core size 153 of 198  vol = 188.539
core size 152 of 198  vol = 182.262
core size 151 of 198  vol = 176.954
core size 150 of 198  vol = 170.712
core size 149 of 198  vol = 166.119
core size 148 of 198  vol = 159.796
core size 147 of 198  vol = 153.767
core size 146 of 198  vol = 149.092
```

```
core size 145 of 198  vol = 143.657
core size 144 of 198  vol = 137.138
core size 143 of 198  vol = 132.517
core size 142 of 198  vol = 127.231
core size 141 of 198  vol = 121.574
core size 140 of 198  vol = 116.775
core size 139 of 198  vol = 112.57
core size 138 of 198  vol = 108.17
core size 137 of 198  vol = 105.133
core size 136 of 198  vol = 101.249
core size 135 of 198  vol = 97.374
core size 134 of 198  vol = 92.974
core size 133 of 198  vol = 88.184
core size 132 of 198  vol = 84.029
core size 131 of 198  vol = 81.898
core size 130 of 198  vol = 78.019
core size 129 of 198  vol = 75.272
core size 128 of 198  vol = 73.052
core size 127 of 198  vol = 70.695
core size 126 of 198  vol = 68.975
core size 125 of 198  vol = 66.694
core size 124 of 198  vol = 64.394
core size 123 of 198  vol = 62.092
core size 122 of 198  vol = 59.045
core size 121 of 198  vol = 56.629
core size 120 of 198  vol = 54.016
core size 119 of 198  vol = 51.806
core size 118 of 198  vol = 49.652
core size 117 of 198  vol = 48.193
core size 116 of 198  vol = 46.648
core size 115 of 198  vol = 44.752
core size 114 of 198  vol = 43.292
core size 113 of 198  vol = 41.093
core size 112 of 198  vol = 39.147
core size 111 of 198  vol = 36.472
core size 110 of 198  vol = 34.117
core size 109 of 198  vol = 31.47
core size 108 of 198  vol = 29.448
core size 107 of 198  vol = 27.325
core size 106 of 198  vol = 25.822
core size 105 of 198  vol = 24.15
core size 104 of 198  vol = 22.648
core size 103 of 198  vol = 21.069
```

```
 core size 102 of 198  vol = 19.953
 core size 101 of 198  vol = 18.3
 core size 100 of 198  vol = 15.723
 core size 99 of 198  vol = 14.841
 core size 98 of 198  vol = 11.646
 core size 97 of 198  vol = 9.434
 core size 96 of 198  vol = 7.354
 core size 95 of 198  vol = 6.179
 core size 94 of 198  vol = 5.666
 core size 93 of 198  vol = 4.705
 core size 92 of 198  vol = 3.665
 core size 91 of 198  vol = 2.77
 core size 90 of 198  vol = 2.151
 core size 89 of 198  vol = 1.715
 core size 88 of 198  vol = 1.15
 core size 87 of 198  vol = 0.874
 core size 86 of 198  vol = 0.685
 core size 85 of 198  vol = 0.528
 core size 84 of 198  vol = 0.37
 FINISHED: Min vol ( 0.5 ) reached
```

```r
# Use the core atom positions for a more suitable superposition:
core.inds <- print(core,vol=0.5)
```

```
# 85 positions (cumulative volume <= 0.5 Angstrom^3)
  start end length
1     9  49     41
2    52  95     44
```

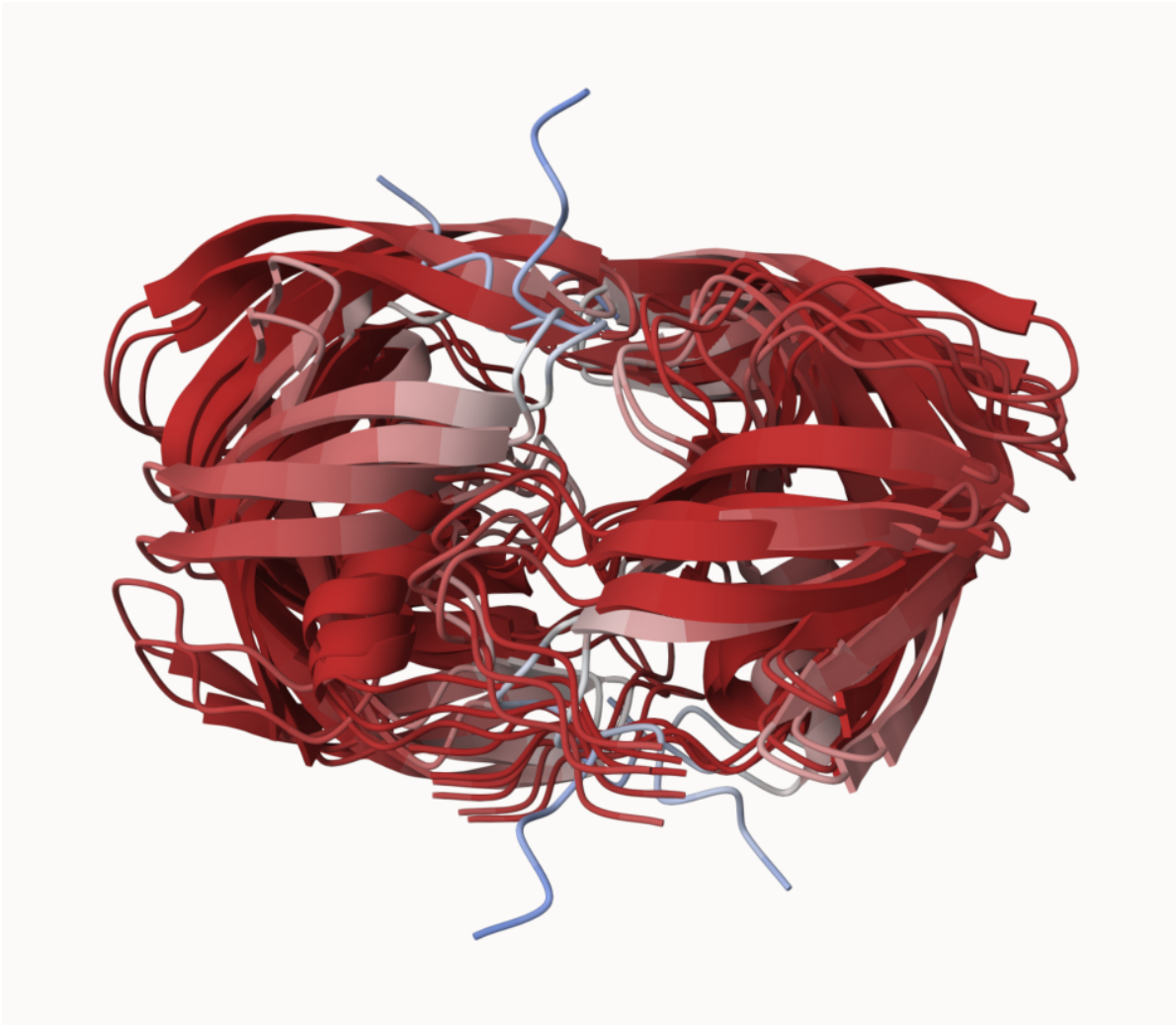


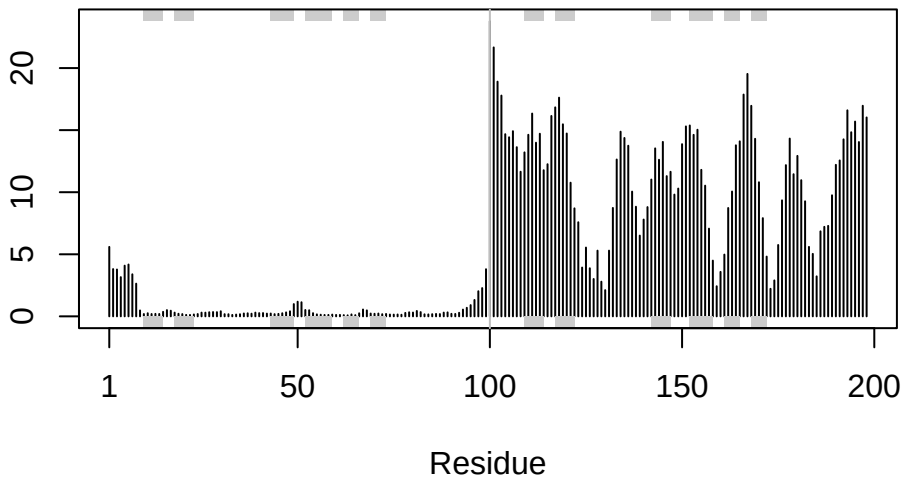
Figure 3: Resulting superposed coordinates with color of atom property set to "Uncertainty/Disorder"

Now we look at RMSF between positions of the structure:

```
xyz <- pdbfit(pdbs, core.inds, outpath="corefit_structures")
```

```
rf <- rmsf(xyz)

plotb3(rf, sse=pdb)
abline(v=100, col="gray", ylab="RMSF")
```



Predicted Alignment Error (PAE) for Domains

AlphaFold produces PAE output, detailed in JSON format (one for each structure)

```
library(jsonlite)

# Listing of all PAE JSON files
pae_files <- list.files(path=results_dir,
                        pattern=".*model.*\\.json",
                        full.names = TRUE)
```

Let's read the first and fifth files:

```
pae1 <- read_json(pae_files[1],simplifyVector = TRUE)
pae5 <- read_json(pae_files[5],simplifyVector = TRUE)

attributes(pae1)
```

```
$names
[1] "plddt"   "max_pae" "pae"     "ptm"     "iptm"
```

```r
# Per-residue pLDDT scores
#  same as B-factor of PDB..
head(pae1$plddt)
```

```
[1] 90.81 93.25 93.69 92.88 95.25 89.44
```

The lower the max PAE score the better…

```r
pae1$max_pae
```

```
[1] 12.84375
```

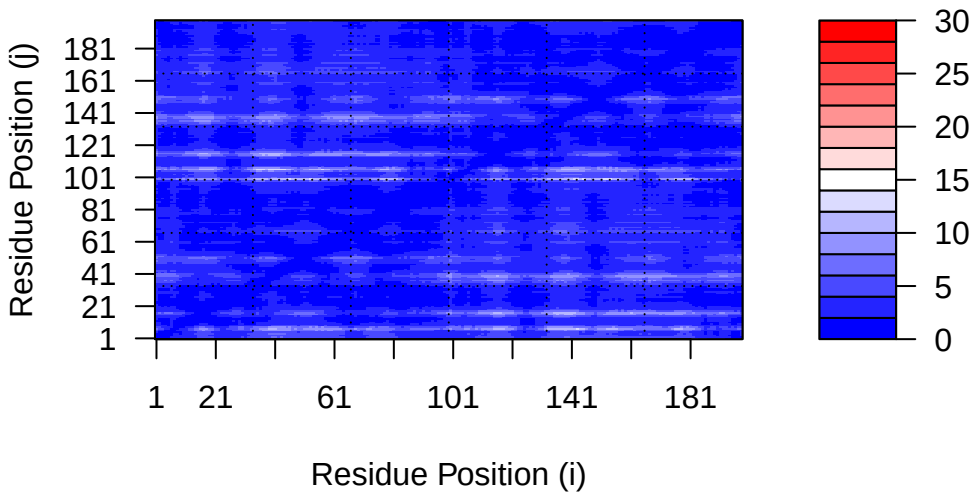
```r
pae5$max_pae
```

```
[1] 29.59375
```

Plot N by N (number of residues)

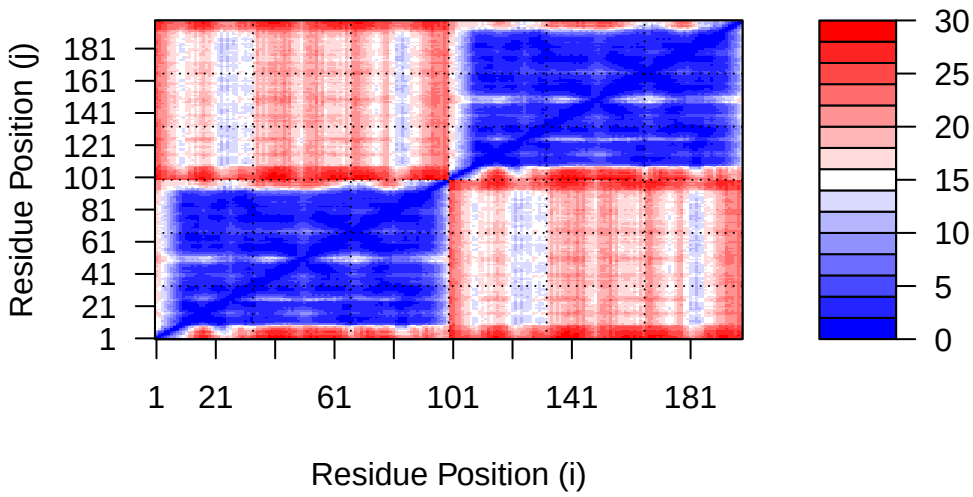
Plot for Model 1:

```r
plot.dmat(pae1$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)",
          grid.col = "black",
          zlim=c(0,30))
```



Plot for Model 5:

```
plot.dmat(pae5$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)",
          grid.col = "black",
          zlim=c(0,30))
```



Residue Conservation from Alignment File

```
aln_file <- list.files(path=results_dir,
                       pattern=".a3m$",
                        full.names = TRUE)
aln_file
```

```
[1] "HIVpr_23119//HIVpr_23119.a3m"
```

```
aln <- read.fasta(aln_file[1], to.upper = TRUE)
```

```
[1] " ** Duplicated sequence id's: 101 **"
[2] " ** Duplicated sequence id's: 101 **"
```

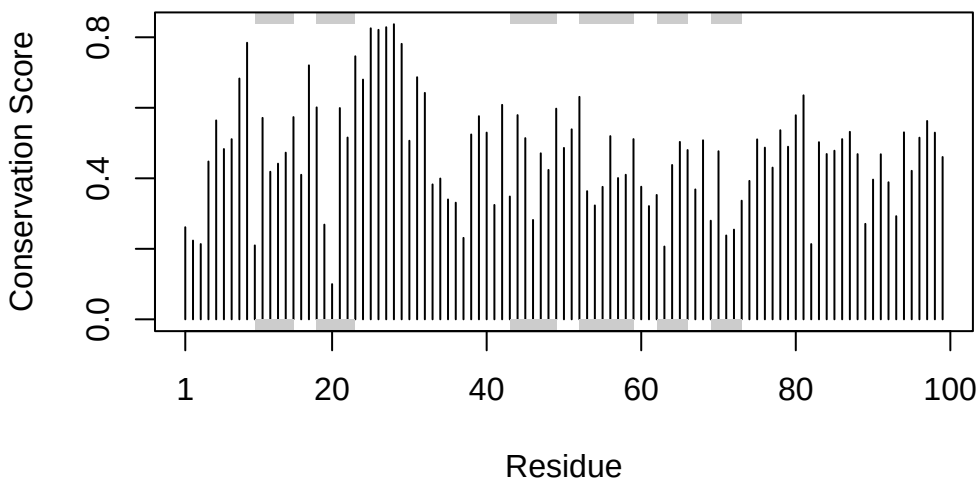
Number of sequences in the alignment:

```
dim(aln$ali)
```

```
[1] 5397  132
```

Score the residue conservation:

```
sim <- conserv(aln)
plotb3(sim[1:99], sse=trim.pdb(pdb, chain="A"),
       ylab="Conservation Score")
```



Conserved active site residues D25, T26, G27, and A28 confirmed:

```
con <- consensus(aln, cutoff = 0.9)
con$seq
```

```
  [1] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [19] "-" "-" "-" "-" "-" "-" "D" "T" "G" "A" "-" "-" "-" "-" "-" "-" "-" "-"
 [37] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [55] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [73] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [91] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[109] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[127] "-" "-" "-" "-" "-" "-"
```